OmniMed-FL: A Robust Multimodal Federated Learning Framework for Clinical Diagnosis

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***Abstract*—Clinical assessment often combines medical images with accompanying text, while privacy and governance constraints can limit centralized data pooling. We present OmniMed-FL, a controlled systems study of multimodal federated learning for five-class clinical condition classification (Normal, Pneumonia, COVID-19, Pleural Effusion, Cardiomegaly). Our proxy corpus pairs 2,400 public chest radiographs and 600 synthetic COVID-19 surrogates with 3,000 class-conditioned synthetic notes, matched by class, not by patient. The framework benchmarks eight fusion strategies, three initializations, four missing-text imputation rules, and matched federated baselines under non-IID Dirichlet partitioning across 3 to 20 hospital clients. Because all notes and the COVID-19 images are synthetic and pairing is not patient-level, these are descriptive proxy comparisons, not estimates of diagnostic performance or deployment readiness. Within those limits: at $K{=}5$ and severe skew ($\alpha{=}0.1$), local-only training achieves a macro-F1 score of 0.310, FedAvg achieves $0.652{\pm}0.057$, FedProx $0.685{\pm}0.064$, a matched FedMME-style one-shot ensemble $0.537{\pm}0.217$, and our SCAFFOLD–AdamW adaptation $0.192{\pm}0.129$, the 0.033 FedProx–FedAvg gap falling inside either two-seed standard deviation. Over a $4{\times}3$ grid, label skew costs up to 0.28 F1 (0.874–0.896 at $\alpha{=}5$ against 0.615–0.741 at $\alpha{=}0.1$) whereas a near-sevenfold client increase costs at most 0.13, while bidirectional volume grows linearly to 183.5 GiB at $K{=}20$. A branch audit is starker: multimodal concatenation spends $2.3\times$ the model state of a text-only branch scoring 0.111 F1 higher.**

***Index Terms*—Multimodal Federated Learning, Vision–Language Models, Clinical Condition Classification, E-Health, Non-Identically Distributed Data, Communication Cost, Retrieval**

I. Introduction

CLINICAL prediction improves when a radiograph is read alongside the text filed with it, and a centralized model can do that, provided somebody is permitted to pool the data. Federated learning (FL) avoids the pooling step by exchanging parameter updates instead of examples [1]–[3], although keeping raw records at home is a data-locality property, not a privacy proof.

Federated medical imaging is well trodden by now [4]–[6], and so is centralized medical vision–language modeling. Their intersection is not. Put the two together and four questions arrive at once: how badly does label skew hurt, when does a five-class head collapse onto its majority class, which fusion operator earns its compute, and what does a multimodal model cost to move. OmniMed-FL answers those four for a controlled proxy benchmark. Our radiographs and synthetic notes are not patient matched, so we measure the learning system, not clinical diagnosis. "Robust" here means stress tests across partitions, client counts, and missing modalities, not a clinical guarantee.

A. Contribution

- Our contribution is a protocol, not another fusion operator. Heterogeneity, client count, anti-collapse components, initialization, and eight fusion rules all run inside one federated loop that fixes the corpus, partition draw, shard profile, encoders, optimizer, and local budget, so a difference between arms is a difference in the named factor and nothing else.
- We add matched controls for two recent multimodal FL methods: a FedMME-style one-shot ensemble, at both our 24-epoch and FedMME's native 100-epoch budget, and a P-FIN-style missing-text stress test over four imputation rules.
- We report the systems side that accuracy tables usually omit: realized per-client skew, GPU-synchronized round time, peak memory, formula-derived bidirectional FP32 volume, and retrieval over all 600 validation queries.
- We report unresolved comparisons explicitly. The fusion sweep, the FedProx–FedAvg gap, and most anti-collapse contrasts at moderate skew are smaller than their two-seed spreads and therefore do not support a reliable ranking.

II. Related Work

A. Medical Image Classification

Early automated diagnostics treated the chest radiograph as a pure computer-vision problem. CheXNet and CheXpert set the centralized CNN baselines [1], [2], and Vision Transformers later showed that global context helps on the same task [7], [8]. All of them pool data on one server, and none reads the note a clinician wrote next to the image.

B. Federated Clinical Learning

To stay inside privacy regulation, the clinical community moved to FL: FedAvg for decentralized training [3], multi-institutional collaboration without sharing patient records [4], multinational COVID-19 CT validation [5], and privacy-enhanced image learning [6]. FedProx and SCAFFOLD attack the client heterogeneity that degrades naive averaging [9], [10]; clinical variants add prompt aggregation, split hybrids, architecture search, personalization, label-scarce transfer, distillation, foundation models, and hardened privacy [22], [23], [25], [26], [28]–[33], [35]–[37]. These frameworks train across distributed hospitals, but stay strictly unimodal.

C. Vision–Language Models in Healthcare

Concurrently, BiomedCLIP, LLaVA-Med, and MedCLIP aligned radiographs with reports and demonstrated what fusion buys [11]–[13], on centralized corpora no single hospital can assemble. Multimodal FL is only now catching up: FedMME votes over one-shot client ensembles, and P-FIN imputes missing features with calibrated uncertainty [14], [15], and federated clinical VLMs are only now appearing [24], [27], [34].

Synthesis. Existing combinations use different datasets, backbones, and budgets, making their headline scores incomparable. We therefore rebuild FedMME- and P-FIN-style arms inside our own loop as matched adaptations rather than comparing scores across papers. Holding everything but one factor fixed enables attribution: it is why we can attribute the one-shot deficit to aggregation rather than to FedMME's larger local budget, and why an eight-way fusion sweep that looks decisive turns out to be noise.

III. System Design

A. Architecture

Figure 1 lays out the pipeline. We tokenize notes with WordPiece at a 128-token cap, resize radiographs to 224×224, and normalize them against ImageNet statistics. The DistilBERT [16] `[CLS]` vector is projected to $\mathbf{h}_t \in \mathbb{R}^{256}$, and mean-pooled ViT tokens give $\mathbf{h}_v \in \mathbb{R}^{512}$. Eight lightweight operators then merge the two: projected concatenation, residual cross-attention, gated residual fusion, dual-projection concatenation, a one-layer decoder A, a distinct two-query/two-layer decoder B, non-residual 384-dimensional cross-attention, and a two-token Transformer encoder. Some code keys still carry foundation-model names from an earlier iteration; they are compact fusion rules, not implementations of those models. All eight feed the same two-layer, five-class multilayer perceptron (MLP) and share the data, partition, optimizer, and local budget. The question we wanted to settle was narrow: can a cheap projection hold its own against heavier attention machinery on hardware a hospital actually owns?

B. Federated and Local Objectives

Let K clients hold disjoint shards $\mathcal{D}_k$ of the training split, with $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$, and let θ collect the trainable parameters on the fusion path: both encoders, the projection layers, the fusion operator, and the classification head. We minimize the sample-weighted empirical risk

$$\begin{aligned} \min_{\theta}\ F(\theta) &= \sum_{k=1}^{K} \frac{n_k}{n} F_k(\theta), \\ F_k(\theta) &= \frac{1}{n_k} \sum_{(x,z,y)\in\mathcal{D}_k} \ell\big(f_\theta(x,z), y\big). \end{aligned} \tag{1}$$

where x is a radiograph, z its class-paired note, y the class label, and f_θ the fused five-class classifier. Shards never leave a client, and the server sees only the trainable tensors listed above. For mini-batch $\mathcal{B}$, the loss we actually implement is

$$\ell_{\mathcal{B}} = \frac{1}{|\mathcal{B}|} \sum_{i\in\mathcal{B}} (1-p_{i,y_i})^{\gamma} \mathrm{CE}_{\epsilon,i} + \lambda\phi(h) + 5[q-0.9]_+, \tag{2}$$

with p_{i,y_i} the true-class probability, $\mathrm{CE}_{\epsilon,i}$ label-smoothed cross-entropy, γ=2, ϵ=0.05, C=5, $h = H(\bar{p})/\log C$, $q = |\mathcal{B}|^{-1}\sum_i \max_c p_{ic}$, $\phi(h) = (1-h)[1-\frac{8}{3}[h-0.7]_+]$, and $[u]_+ = \max(u, 0)$. The entropy-diversity term grows whenever a batch's predictions concentrate, and we attenuate it above normalized entropy 0.7 so it stops pushing once the head is healthy. Setting λ=0 removes that term alone, while the confidence penalty $5[q-0.9]_+$ stays fixed everywhere. The class-balanced sampler and the entropy-diversity term are the two anti-collapse components we ablate in Section IV. Predicted-class diversity is the fraction of the five labels that ever appear as a validation-set argmax.

Each round, the server broadcasts $\theta^{(t)}$, every eligible client runs three local AdamW epochs on class-balanced mini-batches under (2), clips the gradient norm at one, and returns its trainable tensors for the sample-weighted update in (1). Operational runs start from public DistilBERT and ViT-Base/16 weights with random task heads. We fine-tune both encoders and keep unused unimodal heads and the inactive CNN fallback out of optimization and communication.

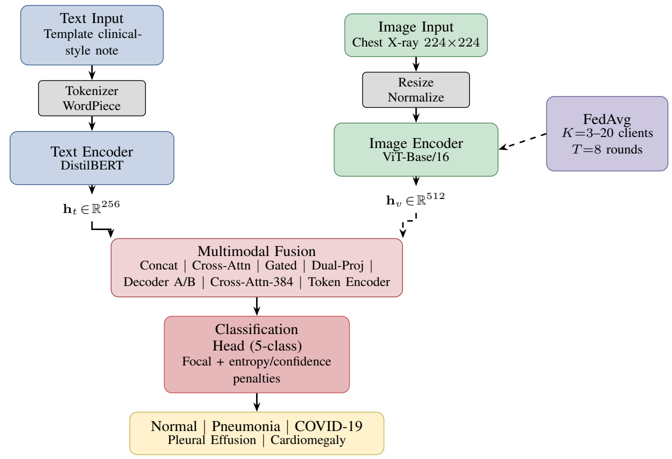

Fig. 1. OmniMed-FL architecture. DistilBERT and ViT-Base/16 encode the two proxy modalities; lightweight multimodal fusion combines them; FedAvg coordinates simulated clients without exchanging their raw examples.

C. Controlled Proxy Corpus

Image data. We built a balanced 3,000-image proxy corpus at 600 examples per class. The public radiographs come from the Kermany pneumonia collection [17] and the NIH ChestX-ray dataset [18], and our loader logs verify 2,400 public images spanning Normal, Pneumonia, Pleural Effusion, and Cardiomegaly. The COVID-19 source we originally specified was unavailable when the runs executed, so the 600 COVID-19 images are synthetic X-ray-style surrogates carrying bilateral opacity patterns, which leaves the image axis partly confounded with source collection and synthetic status. All images are resized to 224×224, ImageNet-normalized, shuffled deterministically, and split 80/20 into 2,400 training and 600 validation examples.

Clinical text data. No patient electronic health record (EHR) data are used. Real EHR text is hard to source publicly, so we wrote a template engine instead. It pairs each image label, at the class level, with a synthetic clinical-style note assembled from observation, symptom, context, and indicator slots. To keep the network from memorizing an obvious keyword we engineered the mix deliberately: 25% of notes draw every slot from the target class, 35% mix target and non-target slots, and 40% are heavily mixed while retaining a probabilistic target-class indicator. This cuts direct label wording, but image and note do not come from the same patient and must never be described as a real image–report pair.

What the corpus does and does not support. Every arm sees the same controlled corpus, which is what makes our internal comparisons meaningful. The absolute scores are another matter. Templates can leave shortcuts, label-level pairing strips out the discordance and missingness of real records, our surrogates may not resemble clinical COVID-19 radiographs, and the split is grouped by neither patient nor source.

D. Experimental Setup

Hyperparameters. Table I collects the protocol. Every client trains with AdamW at learning rate 10^{-4} and weight decay 0.01 for three local epochs per round. The suite runs eight rounds at batch size 16 in FP32, without automatic mixed precision (AMP). Our reference setting is five clients at Dirichlet $\alpha = 1.0$, and Section IV varies α and the client count explicitly. Seeds 0 and 1 drive initialization, sampling, and partitioning, with one exception: pooled-start reuses a single seed-0 checkpoint under seed-specific FL partitions. We ran everything in PyTorch on one NVIDIA H100 NVL device in a server we share with unrelated workloads. The GPU-synchronized timer wraps sequential local training and aggregation within a round, and excludes validation, setup, I/O, networking, concurrency, and multi-node throughput. Our timings are an execution-cost record, not distributed throughput. We also did not force deterministic CUDA algorithms, so a seed fixes the software pseudorandom streams without buying identical reruns.

Partitioning. For each class, a Dirichlet draw allocates training indices across clients. Smaller α concentrates client label distributions, and $\alpha \to \infty$ approaches independent and identically distributed (IID) allocation [19]. A nominal α does not fully describe a finite split, so we report the per-client class shares we actually drew alongside our scalability results. The split is fixed across rounds for a given seed. Shards holding fewer than four examples skip local optimization but still contribute the unchanged global state at their sample weight. We do not renormalize over active clients, which is why our cells report active alongside nominal K.

TABLE I
COMMON SIMULATION PARAMETERS.

Parameter	Value	Parameter	Value
Local Epochs (E)	3	Seeds	0, 1
Fed. Rounds (T)	8	Grid K (ref.)	3, 5, 10, 20 (5)
Batch Size	16	Grid α (ref.)	0.1, 1, 5 (1)
AdamW LR	10^{-4}	K=5 extra α	0.3, 0.5
Weight Decay	0.01	Precision	FP32, no AMP
Diversity wt. (λ)	1.0	Init.	Public enc. + rand. heads

Matched recent-method controls. Our FedMME-style arm uploads each client model once and applies equal-weight hard voting, breaking ties by mean softmax [14]. We run it at two local budgets. Our matched 24 epochs isolates the method from the budget; FedMME's native 100 epochs removes the budget discrepancy. Reporting both keeps the two confounds separate. Either way it keeps the common corpus, shard profile, encoders, and optimizer, substituting our five-class inputs for FedMME's generated-report pipeline. The P-FIN-style stress test fixes three multimodal and two image-only clients, projects both encoders to 256-dimensional normalized features, and compares zero filling, deterministic FIN, Gaussian β-negative-log-likelihood (β-NLL) imputation (β=0.5), and uncertainty-weighted aggregation with Fed-UQ-Avg balance weight 0.6 and temperature 0.2 [15]. We kept concatenation instead of P-FIN's bidirectional cross-modal attention: with one pooled feature per modality, cross-attention carries unit weight and collapses to a linear projection. FedProx uses μ=0.01.

Why these settings. Within an ablation family, only the named factor moves. Three local epochs across eight rounds give a tractable 24-local-epoch budget, and batch size 16 is simply what holds both encoders plus one local model copy in GPU memory for the largest FP32 model. The grid crosses K=$\{3, 5, 10, 20\}$ with α=$\{0.1, 1, 5\}$ and the K=5 sweep adds α=0.3 and 0.5, so the study spans severe to near-IID skew; K=5 is our small cross-silo reference and K=3–20 probes count sensitivity. Two seeds support descriptive claims, not significance. Everything else in Table I is a fixed default, not a per-model optimum.

IV. RESULTS

A. Matched Baselines under Severe Skew

Every arm in Fig. 2(a) shares the corpus, partition draw, encoders, and optimizer at α=0.1, K=5; all but the native-budget FedMME arm also share the 24-local-epoch budget. Local-only training averages a macro-F1 score of 0.310 across the five clients of one partition. Over two matched seeds, FedAvg reaches 0.652±0.057 and FedProx 0.685±0.064. Their

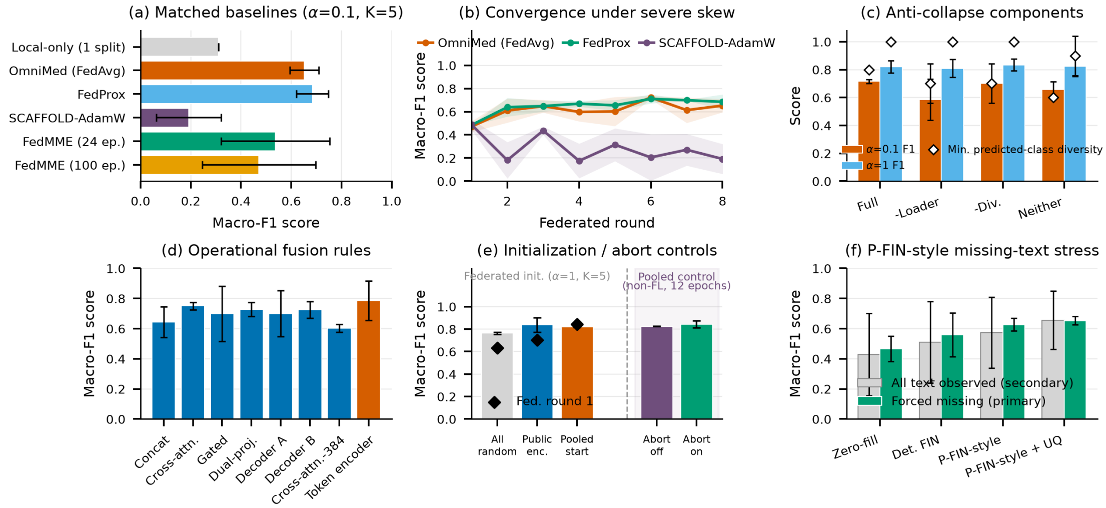

Fig. 2. Matched results (two-seed mean±SD unless noted). (a) Severe-skew baselines; local-only is a five-client mean from one partition, iterative arms use 8 rounds of 3 epochs. (b) Iterative convergence. (c) Anti-collapse F1, diamonds for minimum predicted-class diversity. (d) Eight fusion rules trained in-loop at α=0.1. (e) Federated initialization (round-1 diamonds) and separated non-federated abort controls, whose trigger never fires. (f) P-FIN-style adaptation; held-out text forced missing for the primary metric.

0.033 gap is smaller than either sample standard deviation, so these runs do not separate the two methods. Our SCAFFOLD–AdamW adaptation lands at 0.192 ± 0.129, and the oscillation in panel (b) points at our AdamW control-variate adaptation rather than at classical SGD-based SCAFFOLD. The FedMME-style arm trains five client models independently and combines them by equal-weight hard voting with mean-softmax ties. At our matched 24-epoch budget it reaches 0.537 ± 0.217 (seeds 0.690, 0.384), and at FedMME's native 100-epoch budget it reaches 0.472 ± 0.225 (seeds 0.631, 0.312). Quadrupling local computation does not close the gap to iterative aggregation, so the gap belongs to one-shot aggregation under severe skew, not to a budget handicap. Both budgets are our most seed-sensitive arms for a structural reason: one aggregation has no later round in which to repair a poor local solution. Every arm except SCAFFOLD–AdamW clears local-only training, though the overlapping intervals make that a coarse ordering rather than a ranking.

B. Anti-Collapse, Fusion, and Initialization Ablations

Under severe skew the full class-balanced-sampler and entropy-diversity stack reaches 0.716 ± 0.014 F1 at minimum predicted-class diversity 0.8. Drop the sampler, the entropy-diversity term, or both, and F1 falls to 0.585 ± 0.148, 0.701 ± 0.011, and 0.657 ± 0.057, with minimum diversity 0.7 ± 0.14, 0.7 ± 0.14, and 0.6 ± 0.00 (Fig. 2(c)). The sampler is the larger and more variance-reducing of the two, and only the full stack keeps four of five classes predicted every round. The confidence penalty in (2) stays fixed across all four arms. At α=1 the same four configurations land within 0.809–0.834 with every minimum diversity at or above 0.9. These components earn their place under severe skew and are close to inert at moderate skew. One more observation: three suites nominally repeat the same severe-skew FedAvg/concatenation setting, yet return 0.652 ± 0.057, 0.674 ± 0.185, and 0.716 ± 0.014, differing by more than their own spreads. We report them separately rather than pool them or quote the flattering one.

Panel (d) trains all eight fusion rules inside the operational federated loop. The means run from 0.603 for non-residual 384-dimensional cross-attention up to 0.787 for the two-token Transformer encoder, with per-rule sample standard deviations of 0.024–0.183. Since the widest interval exceeds the entire between-rule spread, the sweep cannot rank fusion operators, and we will not. Projected concatenation, our default everywhere else, sits mid-pack at 0.643 ± 0.102.

Panel (e) separates three federated initializations at α=1. The all-random arm keeps both architectures but initializes every weight from scratch, opening at 0.635 round-1 F1 and closing at 0.765 ± 0.010. Public encoders with random task modules, our operational default, open at 0.704 and close at 0.839 ± 0.065. Pooled-start reuses one seed-0 checkpoint from a 12-epoch centralized run for both FL seeds, so its pretraining sits outside the shared budget; it opens far higher at 0.843 and ends lower at 0.818 ± 0.004. Public encoders buy roughly 0.07 F1 over random initialization; the pooled checkpoint buys a round-1 head start that does not survive to round 8. It was also selected on the reporting validation split, making it a diagnostic, not an unbiased generalization estimate. Non-federated abort-off/on controls post epoch-12 F1 of 0.825 ± 0.001 and 0.843 ± 0.032, best-epoch 0.845 ± 0.013 and 0.884 ± 0.003, on pooled data with sampling and diversity off. All complete 12 epochs at diversity 1.0, so the branch never fires and the gap is rerun variation.

Panel (f) is our matched P-FIN-style adaptation, where 256-dimensional L_2-normalized features and concatenation plus an MLP replace the source bidirectional-attention stack. On the primary forced-missing-text metric the four imputation rules order consistently: zero filling 0.465 ± 0.084, deterministic FIN 0.559 ± 0.144, Gaussian β-NLL imputation 0.627 ± 0.042, and uncertainty-weighted aggregation 0.653 ± 0.027. The ordering

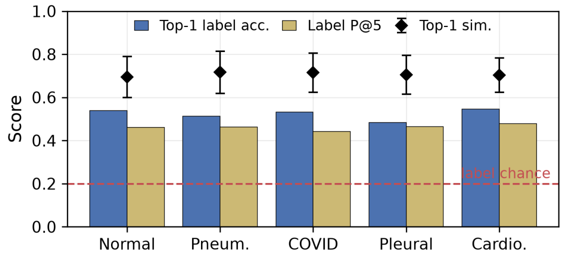

Fig. 3. FAISS exact inner-product retrieval over 2,400 indexed training-note vectors for 600 validation-note queries; TF–IDF is fitted on training notes only. Bars show top-1 same-label accuracy and same-label precision@5; diamonds show mean top-1 cosine similarity with population-SD whiskers. The dashed 0.2 line is a five-class label prior, not a similarity baseline; this evaluates retrieval, not generation.

holds in both seeds, and the spread tightens as the method models its own uncertainty, the qualitative behavior P-FIN reports. The secondary all-text-observed scores are far noisier, 0.429–0.657 with SD up to 0.274, so missing text is the better-determined comparison.

C. Retrieval-Stage Evaluation for Retrieval-Augmented Generation (RAG)

Exact inner-product FAISS search [20] over training-fitted TF–IDF note vectors gives condition-macro averages of 0.522 for top-1 same-label retrieval accuracy, 0.459 for same-label precision@5, and 0.707 for mean top-1 cosine similarity, with the five condition rows covering all 600 validation queries (Fig. 3). Top-1 accuracy sits well above the 0.2 five-class label prior yet far below the 0.707 mean similarity. That combination is the signature of a template corpus: neighbors come out lexically close whether or not they share a label. This check is neither a fine-tuned DistilBERT result nor a generated-answer evaluation, and it establishes no clinical retrieval quality.

D. Scalability, Communication, and Resource Cost

Figure 4(a) assembles the full Cartesian product of $K=\{3,5,10,20\}$ and $\alpha=\{0.1,1,5\}$, and one factor dominates throughout: label skew, not client count. Every $\alpha=5$ cell lies in 0.874–0.896 and every $\alpha=1$ cell in 0.812–0.863, whereas the $\alpha=0.1$ row drops to 0.615–0.741 with far wider spreads, up to ± 0.185 at $K=5$. Within a row, moving K by a factor of nearly seven shifts the mean by at most 0.13 F1, and the $\alpha=0.1$ row is not even monotone in K. Our auxiliary $K=5$ checks at $\alpha=0.3$ and 0.5 give 0.769 ± 0.043 and 0.824 ± 0.113. Cells report A/K, where A counts clients whose realized shard reaches four samples: allocation is complete except under severe skew, where $K=10$ retains 9–10 and $K=20$ retains 18–20 active clients. Panel (f) shows the realized per-client class counts behind one severe-skew cell, where single clients hold entire classes.

Panels (b) and (c) report runtime and peak allocated memory for the timed block on one shared H100 NVL. Timed sections span 42–114 s per round and peak memory 4.77–5.92 GiB. Neither panel is a scaling curve: the $K=5$ column and the $\alpha=1$ row are inherited from our earlier validated suite and ran in a different session, which produces the 5.92 GiB cells and the slowest timed sections. Sequential simulation and contention, not client count, explain the rest.

For $|\theta|$ trainable parameters, b bytes/parameter, T rounds, and nominal K, the plotted bidirectional volume is

$$V_{\mathrm{nom}} = 2KT|\theta|b, \tag{3}$$

Here $|\theta|=153{,}935{,}621$, $b=4$, and $T=8$, so a 615,742,484-byte model state gives nominal bidirectional volumes of 27.5/45.9/91.8/183.5 GiB at $K=3/5/10/20$ (1 GiB= 2^{30} bytes). We keep the nominal count when $A<K$; swapping A for K is an alternative estimate, not a measurement. Serialization, headers, retransmission, secure aggregation, compression, and algorithm-specific state all sit outside it. Panel (d) puts the operating point plainly. Communication grows linearly in K while accuracy stays nearly flat in K. At $\alpha=5$, going from $K=3$ to $K=20$ buys a 0.014 F1 change for 6.7× the traffic.

The one-run branch audit in panel (e) quantifies the multimodal overhead. Text, image, and multimodal F1 are 0.893/0.740/0.782, one-way FP32 model-state sizes are 0.248/0.324/0.573 GiB, sums of timed round durations are 18.5/33.6/45.0 min, and peak allocated memory is 1.69/3.18/4.77 GiB. Multimodal concatenation costs 2.3× the model state, 2.4× the wall time, and 2.8× the memory of the text branch, and it scores 0.111 F1 below that branch. One run per branch cannot support a blanket claim of multimodal superiority. A plausible explanation is that the text encoder directly exploits class-conditioned template shortcuts.

V. Discussion and Limitations

Partition sensitivity is the most consistent finding: at $\alpha=0.1$, even identically configured FedAvg reruns vary substantially. Three limitations bound its practical interpretation. First, our client-count study is a sequential simulation on a shared server. It models neither concurrent workers, latency, stragglers, and dropout nor heterogeneous accelerators, and we never test aggregation against poisoned or Byzantine updates. Our own dispersion hints at why that is hard: at $\alpha=0.1$ honest updates already spread by ± 0.185 F1, so a trimmed-mean or median aggregator must discard genuine minority-class signal to reject an adversarial one, while sample-weighted FedAvg leaves a large shard unbounded leverage. Skew and Byzantine tolerance pull against each other, and quantifying that needs a threat model we did not build. Our title's "robust" denotes these stress tests, nothing stronger. Second, communication. Eq. (3) exposes model-state scaling and Fig. 4(e) the multimodal footprint, but a real study would measure end-to-end traffic and latency under compression, partial participation, and failures. We implement neither secure aggregation nor the differentially private training of Abadi *et al.* [21]. Keeping raw examples local is a locality property, not a privacy guarantee.

Third, clinical interpretation stays bounded by Section III-C. Neither our absolute F1 nor our retrieval scores estimate diagnostic safety, generalization, clinical utility, or deployment readiness, and with two seeds our spreads are descriptive rather than significant. What we want next is not a longer version of this study but a different one: larger multi-site datasets, more

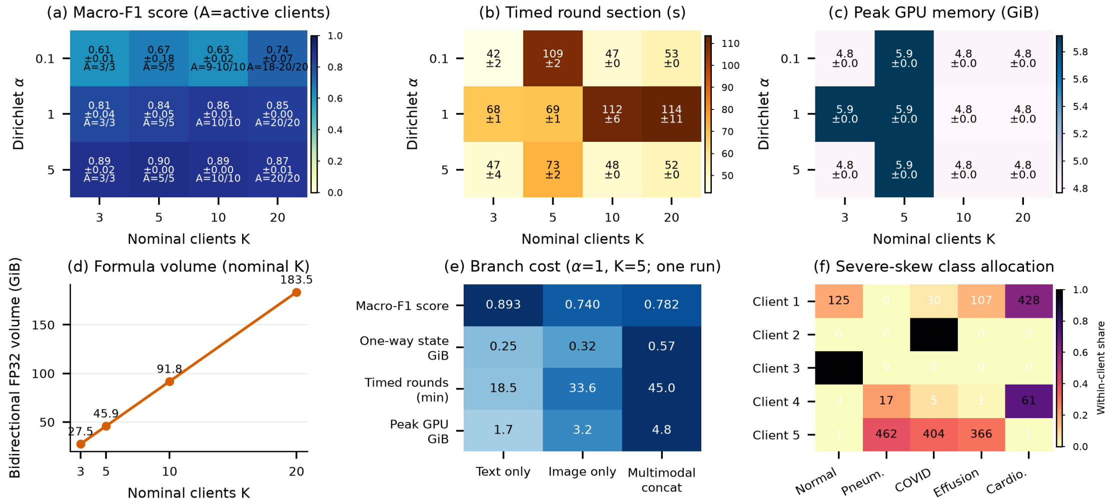

Fig. 4. Scalability/resource audit. Two-seed mean±SD for (a) macro-F1 score (A/K is active/nominal clients), (b) the GPU-synchronized training/aggregation section per round, and (c) peak allocated memory within it. (d) Formula FP32 volume over eight rounds, skipped shards included; not measured traffic. (e) One-run branch cost at $\alpha{=}1, K{=}5$. (f) Severe-skew seed-0 counts.

seeds, native-method replications, calibrated uncertainty, and privacy testing.

VI. Conclusion

OmniMed-FL is a controlled systems study of multimodal FL, not a clinical validation. Across a 4×3 client-count/skew grid, eight fusion rules, three initializations, four missing-text imputation rules, and matched FedAvg, FedProx, SCAFFOLD–AdamW, and FedMME-style baselines, one result holds consistently. Label skew dominates. Client count is near-neutral for accuracy and linear in communication. Everything else is conditional on the partition realization and our protocol choices. Real patient-paired data, external validation, and appropriate safeguards are required before clinical use.

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